# A Short Note on Sums

## 1 A closed form

The sum of the first $n$ positive integers has a closed form. Pairing the first and last terms shows that every pair contributes $n + 1$.

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2} \tag{1}$$

Equation (1) is often attributed to Gauss. A related identity for squares is

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

## 2 Euler's identity

For real $\theta$ we have $e^{i\theta} = \cos\theta + i\sin\theta$, and therefore $e^{i\pi} + 1 = 0$.